



Blacktown Boys' High School

2024

HSC Trial Examination

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided for this paper
- All diagrams are not drawn to scale
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

**Total marks:
70****Section I – 10 marks** (pages 3 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 8 – 15)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Assessor: X. Chirgwin

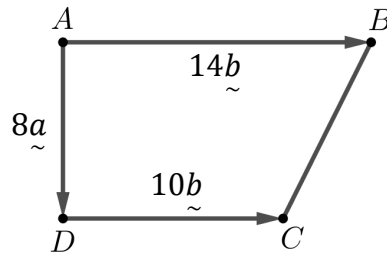
Student Name: _____

Teacher Name: _____

Students are advised that this is a trial examination only and cannot in any way guarantee the content or format of the 2024 Higher School Certificate Examination.

Section I**10 marks****Attempt Questions 1–10****Allow about 15 minutes for this section**

Use the multiple choice answer sheet for Questions 1–10.

Q1. Which of the following is the correct vector of $\overrightarrow{BC}$?

- A. $-8\tilde{a} - 4\tilde{b}$
- B. $-8\tilde{a} + 4\tilde{b}$
- C. $8\tilde{a} - 4\tilde{b}$
- D. $8\tilde{a} + 4\tilde{b}$

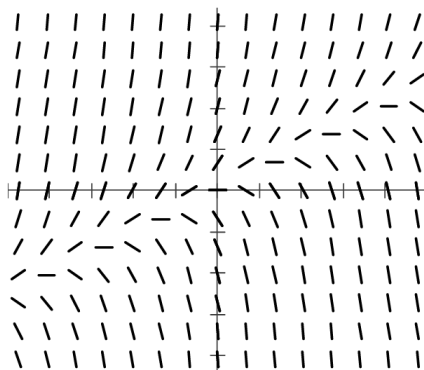
Q2. A standard six-sided die is rolled 18 times.

Let $\hat{p}$ be the proportion of the rolls with an outcome of 3.

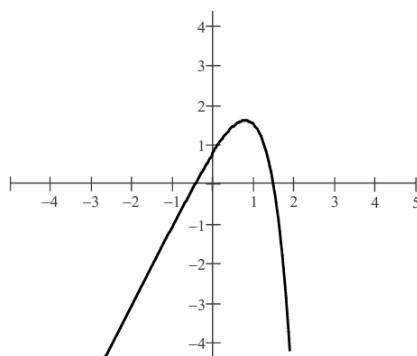
Which of the following expressions is the probability that at most 12 of the rolls have an outcome of 3?

- A. $P\left(\hat{p} \leq \frac{1}{6}\right)$
- B. $P\left(\hat{p} \leq \frac{2}{3}\right)$
- C. $P\left(\hat{p} \geq \frac{1}{6}\right)$
- D. $P\left(\hat{p} \geq \frac{2}{3}\right)$

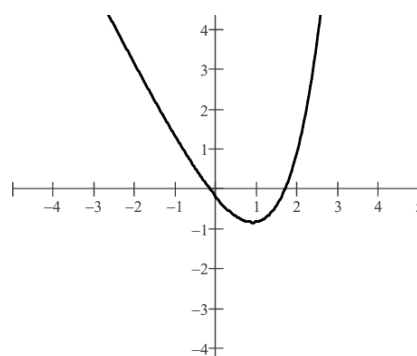
- Q3. Which of the following could be the graph of the solution of the differential equation shown?



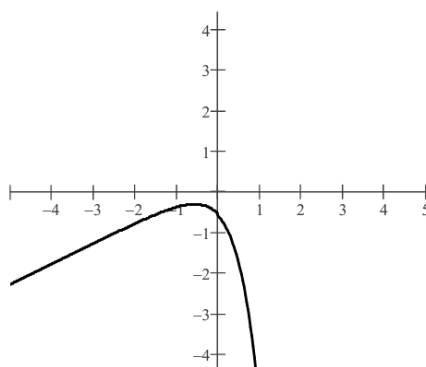
A.



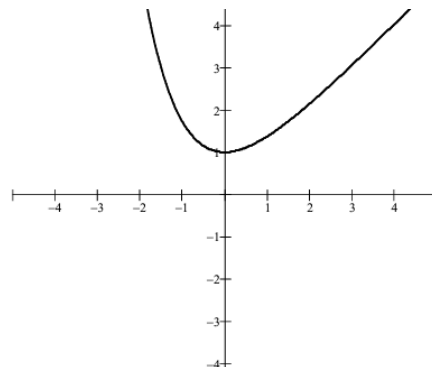
B.



C.



D.



- Q4. A graph has parametric equations $x = 2 \cos t$, $y = 2 - 4 \sin^2 t$. What is its Cartesian equation?

- A. $y = x^2 - 2$ for $-2 \leq x \leq 2$
- B. $y = x^2 - 2$ for $-2 \leq x \leq 0$
- C. $y = x^2 + 2$ for $-2 \leq x \leq 2$
- D. $y = x^2 + 2$ for $-2 \leq x \leq 0$

Q5. If $\sin x = 0.28$ and $\frac{\pi}{2} \leq x \leq \pi$, evaluate $\tan 2x$.

- A. $\frac{527}{336}$
- B. $-\frac{527}{336}$
- C. $\frac{336}{527}$
- D. $-\frac{336}{527}$

Q6. Mohammad hits the target on average 2 out of every 3 shots in archery competitions. During a competition he has 10 shots at the target. What is the probability that Mohammad hits the target exactly 9 times?

- A. $10 \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^9$
- B. $\left(\frac{2}{3}\right)^9 \left(\frac{1}{3}\right)$
- C. $10 \left(\frac{2}{3}\right)^9$
- D. $5 \left(\frac{2}{3}\right)^{10}$

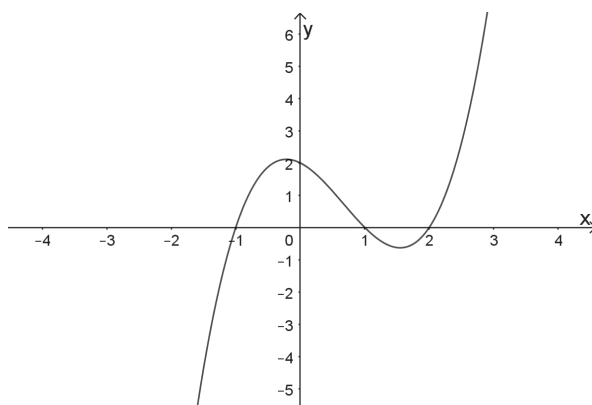
Q7. What is the value of $\tan \alpha$ when the expression $4 \sin x - 3 \cos x$ is written in the form $5 \sin(x - \alpha)$?

- A. $-\frac{3}{4}$
- B. $\frac{3}{4}$
- C. $-\frac{4}{3}$
- D. $\frac{4}{3}$

Q8. What is the range of the function $f(x) = \cos^{-1}(\tan x)$?

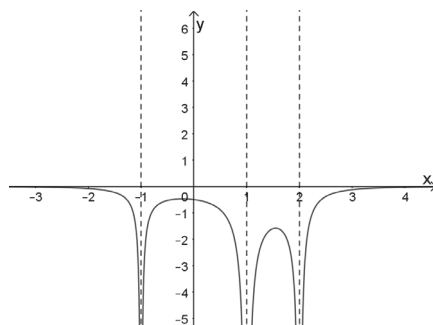
- A. $[0, \pi]$
- B. $(0, \pi)$
- C. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- D. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Q9. The graph of the function $y = f(x)$ is below.

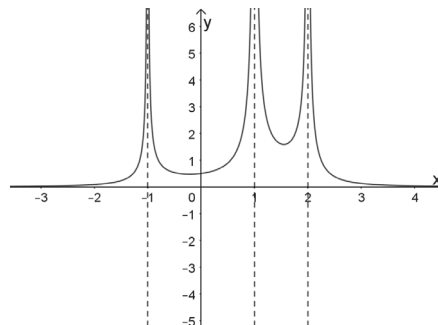


Which of the following is a graph of $y = \frac{-1}{f(|x|)}$?

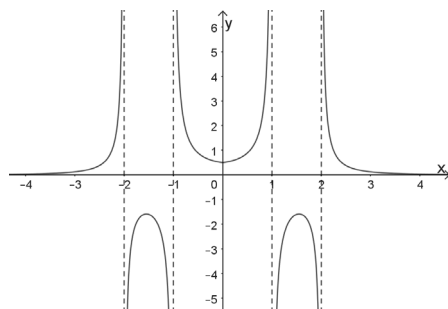
A.



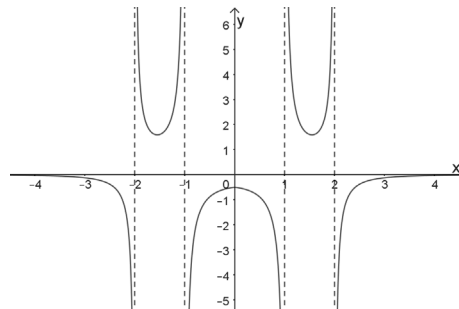
B.



C.



D.



- Q10. Given that $\vec{a}$ and $\vec{b}$ are two non-zero vectors, let $\vec{c}$ be the projection of $\vec{a}$ onto $\vec{b}$.
What is the projection of $12\vec{a}$ onto $3\vec{b}$?

- A. $36\vec{c}$
- B. $12\vec{c}$
- C. $4\vec{c}$
- D. $3\vec{c}$

End of Section I

Section II**60 Marks****Attempt Questions 11–14****Allow about 1 hour and 45 minutes for this section**

Answer each question on a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

(a) For the vectors $\underline{u} = 5\underline{i} + \underline{j}$ and $\underline{v} = 2\underline{i} - 3\underline{j}$, evaluate each of the following.

(i) $2\underline{u} - 3\underline{v}$ 1

(ii) $\underline{u} \cdot \underline{v}$ 1

(b) The polynomial $P(x) = 5x^3 - 2x + 20$ has roots α, β and γ . Evaluate:

(i) $\alpha\beta\gamma$ 1

(ii) $\alpha^2 + \beta^2 + \gamma^2$ 2

(c) Solve $\frac{10x}{1+3x} \leq 3$ 3

(d) Show that $\frac{d}{dx} \left(\frac{\sin^{-1}(3x)}{3x} \right) = \frac{3x - \sin^{-1}(3x) \sqrt{1-9x^2}}{3x^2 \sqrt{1-9x^2}}$ 3

Question 11 continues on next page

Question 11 (continued)

- (e) A recent census found that 55% of people used public transport.

A sample of 600 randomly selected Australians was surveyed.

Let $\hat{p}$ be the sample proportion of surveyed people who were born overseas.

A normal distribution is to be used to approximate $P(\hat{p} \leq 0.575)$.

- (i) Show that the variance of the random variable $\hat{p}$ is $\frac{33}{80000}$. 2
- (ii) Use the standard normal distribution and the information on page 15 to approximate $P(\hat{p} \leq 0.575)$, giving your answer correct to two decimal places. 2

End of Questions 11

Question 12 (15 marks) Use the Question 12 Writing Booklet.

(a) Evaluate $\int_0^{\frac{1}{5}} \frac{dx}{1 + 25x^2}$ 2

(b) Evaluate $\int_{-1}^4 \frac{t}{\sqrt{5+t}} dt$ by using the substitution $t = u - 5$. 3

- (c) Samirali turns on his radio 30 times a month on average. It can receive 20 radio stations. Every time he switches on the radio, it goes to a random station. What is the probability that it will be the radio station Samirali wants to listen to upon switching on the radio at least twice in a month? Round your answer to 4 significant figures. 3

- (d) It is known that ${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$ for all integers such that $1 \leq r \leq n - 1$. (Do NOT prove this.) 2

Find ONE possible set of values for p and q such that

$${}^{2026}C_{1924} - {}^{2024}C_{100} - {}^{2024}C_{101} = {}^pC_q$$

- (e) The vectors $\tilde{u} = \begin{pmatrix} -5 \\ m \end{pmatrix}$ and $\tilde{v} = \begin{pmatrix} 3m - 4 \\ 1 - 6m \end{pmatrix}$ are perpendicular. 2

What are the possible values of m ?

- (f) For all integers $n \geq 1$, use mathematical induction to prove that 3

$$\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \frac{5}{3 \times 4 \times 2^3} + \cdots + \frac{n+2}{n(n+1) \times 2^n} = 1 - \frac{1}{(n+1) \times 2^n}$$

End of Questions 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

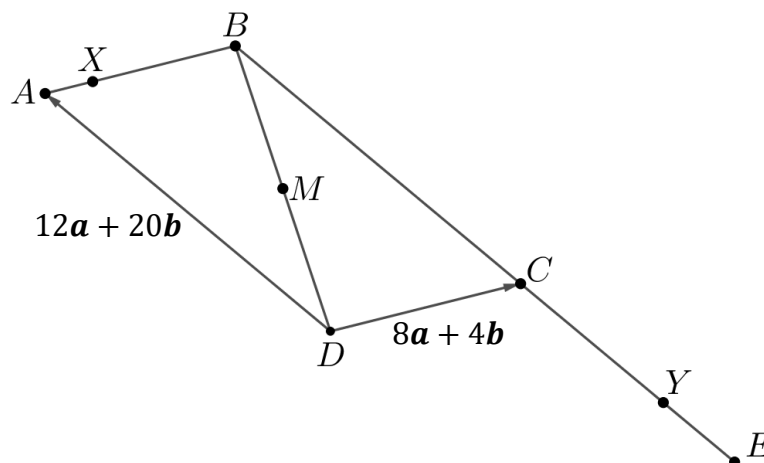
- (a) (i) By expanding the left-hand side, show that 1

$$\sin(8x + 5x) + \sin(8x - 5x) = 2 \sin 8x \cos 5x$$

- (ii) Hence find $\int \sin 8x \cos 5x \, dx$. 2

- (b) The arc of the curve $y = \frac{1}{2}(1 + \sin x)$ between $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{2}$ is 3
rotated about the x -axis. Find the volume of the solid formed.

- (c) $ABCD$ is a parallelogram where $\overrightarrow{DA} = 12\mathbf{a} + 20\mathbf{b}$ and $\overrightarrow{DC} = 8\mathbf{a} + 4\mathbf{b}$.
 X lies on the line $\overrightarrow{AB}$ such that $\overrightarrow{AX} : \overrightarrow{XB} = 1 : 3$. M is the midpoint of $\overrightarrow{DB}$.
 $\overrightarrow{CE}$ is an extension of $\overrightarrow{BC}$. Y lies on the line $\overrightarrow{CE}$ such that $2\overrightarrow{CY} = -\overrightarrow{DA}$.



- (i) Find $\overrightarrow{XM}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. 2
- (ii) Prove that X , M and Y are collinear and find k if $\overrightarrow{XM} = k\overrightarrow{MY}$. 3

Question 13 continues on next page

Question 13 (continued)

(d) (i) Show that $\frac{x+2}{4-x^2} = \frac{1}{2-x}$ **1**

(ii) Find the particular solution to the differential equation **3**

$$2(4-x^2)\frac{dy}{dx} = y(x+2)$$

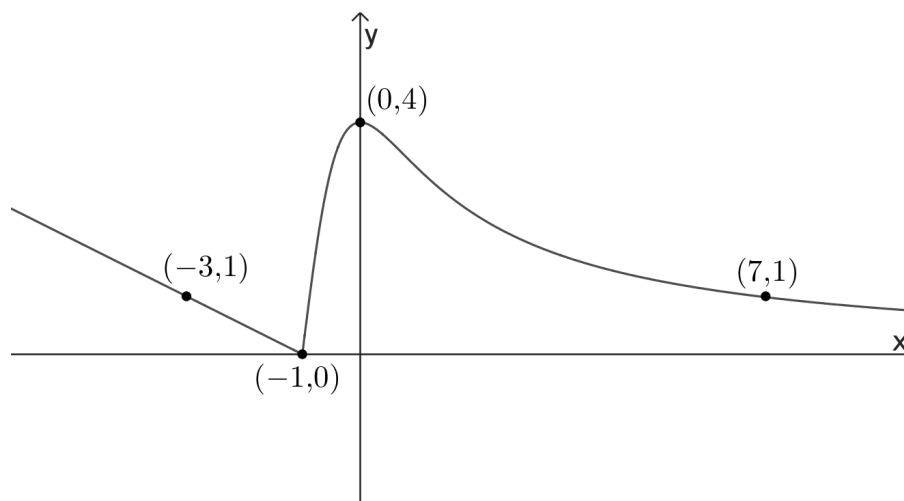
that passes through the point $(1, 2)$, give the answer in the form

$y = f(x)$.

End of Questions 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

- (a) The diagram below shows the graph of the function
- $y = f(x)$
- .

3Sketch $y = \frac{1}{\sqrt{f(x)}}$

- (b) The rate of change of the number of people entering an arena is modelled by $\frac{dN}{dt} = kN \left(1 - \frac{N}{5000}\right)$, where k is a constant and N is the number of people after t minutes. There are 200 people in the arena initially and the capacity of the arena is 5000 people. At a certain time, the number of people in the arena is 1000 and is increasing at the rate of 500 people per minute.

- (i) Show that $k = 0.625$. **1**
- (ii) Show that $\frac{5000}{N(5000 - N)} = \frac{1}{N} + \frac{1}{5000 - N}$ **1**
- (iii) Find the number of people in the arena after 6 minutes. **4**
- (iv) How long will it take for the arena to be 90% full? Round to the nearest second. **1**

Question 14 continues on next page

Question 14 (continued)

(c) Given that $\sin^{-1} x$, $\cos^{-1} x$ and $\sin^{-1}(2 - x)$ have values for $0 \leq x \leq \frac{\pi}{2}$.

(i) Show that $\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1$. **3**

(ii) Hence solve the equation $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(2 - x)$. **2**

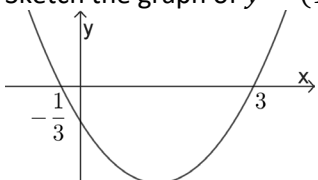
End of Paper

2024 Mathematics Extension 1 AT4 Trial Solutions

Section 1

Q1	C $\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DC}$ $\overrightarrow{BC} = -14\underset{\sim}{b} + 8\underset{\sim}{a} + 10\underset{\sim}{b}$ $\overrightarrow{BC} = 8\underset{\sim}{a} - 4\underset{\sim}{b}$	1 Mark
Q2	B Having 12 out of 18 rolls be an outcome of 3 corresponding to a $\hat{p}$ value of $\frac{12}{18} = \frac{2}{3}$ Since “at most” that proportion is desired, then probability is $P\left(\hat{p} \leq \frac{2}{3}\right)$	1 Mark
Q3	C	1 Mark
Q4	A $x = 2 \cos t, \quad -2 \leq x \leq 2$ $x^2 = 4 \cos^2 t$ $y = 2 - 4 \sin^2 t$ $4 \sin^2 t = 2 - y$ $\sin^2 t + \cos^2 t = 1$ $4 \sin^2 t + 4 \cos^2 t = 4$ $2 - y + x^2 = 4$ $y = x^2 - 2$	1 Mark
Q5	D $\sin x = \frac{7}{25}, \quad \frac{\pi}{2} \leq x \leq \pi$ $\tan x = -\frac{7}{24}$ $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ $\tan 2x = \frac{2 \times \left(-\frac{7}{24}\right)}{1 - \left(-\frac{7}{24}\right)^2}$ $\tan 2x = -\frac{336}{527}$	1 Mark
Q6	D ${}^{10}C_9 \left(\frac{2}{3}\right)^9 \left(\frac{1}{3}\right)^1 = 5 \times \left(\frac{2}{3}\right)^{10}$	1 Mark
Q7	B $5 \sin(x - \alpha) = 5 \sin x \cos \alpha - 5 \cos x \sin \alpha$ $5 \cos \alpha = 4 \dots (1)$ $5 \sin \alpha = 3 \dots (2)$ $(2) \div (1)$ $\tan \alpha = \frac{3}{4}$	1 Mark
Q8	A Range is $0 \leq y \leq \pi$	1 Mark

Q9	D $y = \frac{-1}{f(x)}$ Take the graph where x is positive and reflect it along the y-axis. Then take the reciprocal of this graph, generating vertical asymptotes at $x = \pm 1, \pm 2$, and y-intercept becomes $\frac{1}{2}$. Then reflect the resulting graph along the x-axis, and y-intercept becomes $-\frac{1}{2}$.	1 Mark
Q10	B $\text{proj}_{(3\tilde{b})}(12\tilde{a}) = \frac{(12\tilde{a}) \cdot (3\tilde{b})}{ 3\tilde{b} ^2}(3\tilde{b})$ $\text{proj}_{(3\tilde{b})}(12\tilde{a}) = \frac{12(\tilde{a} \cdot \tilde{b})}{ \tilde{b} ^2}(\tilde{b})$ $\text{proj}_{(3\tilde{b})}(12\tilde{a}) = 12\text{proj}_{(\tilde{b})}(\tilde{a})$ $\text{proj}_{(3\tilde{b})}(12\tilde{a}) = 12\tilde{c}$	1 Mark

Section 2		
Q11ai	$2\tilde{u} - 3\tilde{v}$ $= 2(5\tilde{i} + \tilde{j}) - 3(2\tilde{i} - 3\tilde{j})$ $= 10\tilde{i} + 2\tilde{j} - 6\tilde{i} + 9\tilde{j}$ $= 4\tilde{i} + 11\tilde{j}$	1 Mark Correct solution
Q11aii	$(5\tilde{i} + \tilde{j}) \cdot (2\tilde{i} - 3\tilde{j})$ $= 5 \times 2 + 1 \times -3$ $= 7$	1 Mark Correct solution
Q11bi	$P(x) = 5x^3 - 2x + 20$ $\alpha\beta\gamma = -\frac{d}{a} = -\frac{20}{5}$ $\alpha\beta\gamma = -4$	1 Mark Correct solution
Q11bii	$\alpha^2 + \beta^2 + \gamma^2$ $= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$ $= 0 - 2 \times -\frac{2}{5}$ $= \frac{4}{5}$	2 Marks Correct solution 1 Mark Makes significant progress
Q11c	$\frac{10x}{1+3x} \leq 3, \quad x \neq -\frac{1}{3}$ $10x(1+3x) \leq 3(1+3x)^2$ $10x(1+3x) - 3(1+3x)^2 \leq 0$ $(1+3x)[10x - 3(1+3x)] \leq 0$ $(1+3x)(x-3) \leq 0$ Sketch the graph of $y = (1+3x)(x-3)$  $\therefore -\frac{1}{3} < x \leq 3$	3 Marks Correct solution 2 Marks Makes significant progress and identifies 3, $-\frac{1}{3}$ are the two key values 1 Mark Multiplies both sides by the square of denominator
Q11d	$\frac{d}{dx} \left(\frac{\sin^{-1}(3x)}{3x} \right)$ $= \frac{3x \times \frac{3}{\sqrt{1-9x^2}} - 3 \sin^{-1}(3x)}{9x^2}$ $= \frac{3(3x - \sin^{-1}(3x)\sqrt{1-9x^2})}{9x^2\sqrt{1-9x^2}}$ $= \frac{3x - \sin^{-1}(3x)\sqrt{1-9x^2}}{3x^2\sqrt{1-9x^2}}$	3 Marks Correct solution 2 Marks Make significant progress 1 Mark Differentiate $\sin^{-1}(3x)$
Q11ei	Let p be the probability of people who catch public transport $p = 0.55, \quad q = 1 - p = 0.45$ $n = 600$ $\sigma^2 = \frac{0.55 \times 0.45}{600}$ $\sigma^2 = \frac{33}{80000}$	2 Marks Correct solution 1 Mark Obtains correct p and q values

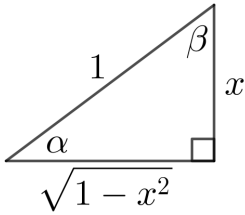
Q11eii	$P(\hat{p} \leq 0.575) = P\left(Z \leq \frac{0.575 - 0.55}{\sqrt{\frac{33}{80000}}}\right)$ $P(\hat{p} \leq 0.575) = P(Z \leq 1.2309 \dots)$ $P(\hat{p} \leq 0.575) \approx P(Z \leq 1.23)$ $P(\hat{p} \leq 0.575) \approx 0.8907$ $P(\hat{p} \leq 0.575) = 0.89 \text{ (2 dp)}$	2 Marks Correct solution 1 Mark Obtains the correct z-score
Q12a	$\int_0^{\frac{1}{5}} \frac{dx}{1 + 25x^2}$ $= \frac{1}{5} \int_0^{\frac{1}{5}} \frac{5dx}{1 + (5x)^2}$ $= \frac{1}{5} [\tan^{-1}(5x)]_0^{\frac{1}{5}}$ $= \frac{1}{5} \left[\tan^{-1}\left(5 \times \frac{1}{5}\right) - \tan^{-1}(5 \times 0) \right]$ $= \frac{1}{5} \left[\frac{\pi}{4} - 0 \right]$ $= \frac{\pi}{20}$	2 Marks Correct solution 1 Mark Correct anti-derivative
Q12b	$I = \int_{-1}^4 \frac{t}{\sqrt{5+t}} dt$ $t = u - 5$ $dt = du$ $t = 4, u = 9$ $t = -1, u = 4$ $I = \int_4^9 \frac{u-5}{\sqrt{(5+u-5)}} du$ $I = \int_4^9 \frac{u-5}{\sqrt{u}} du$ $I = \int_4^9 \left(u^{\frac{1}{2}} - 5u^{-\frac{1}{2}} \right) du$ $I = \left[\frac{u^{\frac{3}{2}}}{\frac{3}{2}} - \frac{5u^{\frac{1}{2}}}{\frac{1}{2}} \right]_4^9$ $I = \left[\frac{2u\sqrt{u}}{3} - 10\sqrt{u} \right]_4^9$ $I = \left[\left(\frac{2 \times 9\sqrt{9}}{3} - 10\sqrt{9} \right) - \left(\frac{2 \times 4\sqrt{4}}{3} - 10\sqrt{4} \right) \right]$ $I = \left[(18 - 30) - \left(\frac{16}{3} - 20 \right) \right]$ $I = \frac{8}{3}$	3 Marks Correct solution 2 Marks Obtains correct anti-derivative in terms of u 1 Mark Obtains correct integrand in terms of u
Q12c	The probability of the correct radio station is $\frac{1}{20}$ $X \sim B\left(30, \frac{1}{20}\right)$ $P(X \geq 2) = 1 - P(X \leq 1)$ $P(X \geq 2) = 1 - (P(X = 1) + P(X = 0))$	3 Marks Correct solution 2 Marks Makes significant progress

	$P(X \geq 2) = 1 - \left({}^{30}C_0 \left(\frac{1}{20} \right)^0 \left(\frac{19}{20} \right)^{30} + {}^{30}C_1 \left(\frac{1}{20} \right)^1 \left(\frac{19}{20} \right)^{29} \right)$ $P(X \geq 2) = 0.446457 \dots$ $P(X \geq 2) = 0.4465 \text{ (4 sig fig)}$	1 Mark Finds $P(X = 0)$ or $P(X = 1)$
Q12d	${}^{2026}C_{1924} - {}^{2024}C_{100} - {}^{2024}C_{101} = {}^pC_q$ ${}^{2026}C_{1924} - ({}^{2024}C_{100} + {}^{2024}C_{101}) = {}^pC_q$ ${}^{2026}C_{1924} - {}^{2025}C_{101} = {}^pC_q$ ${}^pC_q + {}^{2025}C_{101} = {}^{2026}C_{1924}$ ${}^pC_q + {}^{2025}C_{101} = {}^{2026}C_{102}$ ${}^{2025}C_{102} + {}^{2025}C_{101} = {}^{2026}C_{102}$ $\therefore p = 2025, q = 102 \text{ or } 1923$	2 Marks Correct solution 1 Mark Combines two terms
Q12e	$\binom{-5}{m} \cdot \binom{3m-4}{1-6m} = 0$ $-15m + 20 + m - 6m^2 = 0$ $-6m^2 - 14m + 20 = 0$ $3m^2 + 7m - 10 = 0$ $(3m + 10)(m - 1) = 0$ $m = -\frac{10}{3}, m = 1$	2 Marks Correct solution 1 Mark Makes significant progress
Q12f	$RTP: \frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \frac{5}{3 \times 4 \times 2^3} + \dots + \frac{n+2}{n(n+1) \times 2^n}$ $= 1 - \frac{1}{(n+1) \times 2^n}$ 1. Prove statement is true for $n = 1$ $LHS = \frac{3}{1 \times 2 \times 2} = \frac{3}{4}$ $LHS = \frac{3}{4}$ $RHS = 1 - \frac{1}{(1+1) \times 2^1}$ $RHS = 1 - \frac{1}{4}$ $RHS = \frac{3}{4}$ $\therefore \text{statement is true for } n = 1$ 2. Assume statement is true for $n = k$ (k is some positive integer) $\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \frac{5}{3 \times 4 \times 2^3} + \dots + \frac{k+2}{k(k+1) \times 2^k}$ $= 1 - \frac{1}{(k+1) \times 2^k}$ 3. Prove statement is true for $n = k + 1$ $\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \frac{5}{3 \times 4 \times 2^3} + \dots + \frac{k+2}{k(k+1) \times 2^k}$ $+ \frac{k+1+2}{(k+1)(k+1+1) \times 2^{k+1}} = 1 - \frac{1}{(k+1+1) \times 2^{k+1}}$ $\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \frac{5}{3 \times 4 \times 2^3} + \dots + \frac{k+2}{k(k+1) \times 2^k}$ $+ \frac{k+3}{(k+1)(k+2) \times 2^{k+1}} = 1 - \frac{1}{(k+2) \times 2^{k+1}}$	3 Marks Correct solution 2 Marks Makes significant progress 1 Mark Establishes the base case

	$LHS = \frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \frac{5}{3 \times 4 \times 2^3} + \dots + \frac{k+2}{k(k+1) \times 2^k}$ $+ \frac{k+3}{(k+1)(k+2) \times 2^{k+1}}$ $LHS = 1 - \frac{1}{(k+1) \times 2^k} + \frac{k+3}{(k+1)(k+2) \times 2^{k+1}}$ $LHS = 1 - \left(\frac{2(k+2)}{(k+1)(k+2) \times 2^{k+1}} - \frac{(k+3)}{(k+1)(k+2) \times 2^{k+1}} \right)$ $LHS = 1 - \frac{2k+4-k-3}{(k+1)(k+2) \times 2^{k+1}}$ $LHS = 1 - \frac{k+1}{(k+1)(k+2) \times 2^{k+1}}$ $LHS = 1 - \frac{1}{(k+2) \times 2^{k+1}}$ $LHS = RHS$ $\therefore \frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \frac{5}{3 \times 4 \times 2^3} + \dots + \frac{n+2}{n(n+1) \times 2^n}$ $= 1 - \frac{1}{(n+1) \times 2^n}$	
Q13ai	$\sin(8x+5x) + \sin(8x-5x)$ $= \sin 8x \cos 5x + \cos 8x \sin 5x + \sin 8x \cos 5x - \cos 8x \sin 5x$ $= 2 \sin 8x \cos 5x$	1 Mark Correct solution
Q13aii	$\int \sin 8x \cos 5x \, dx$ $= \frac{1}{2} \int (\sin(8x+5x) + \sin(8x-5x)) \, dx$ $= \frac{1}{2} \int (\sin 13x + \sin 3x) \, dx$ $= \frac{1}{2} \times \left(-\frac{1}{13} \cos 13x - \frac{1}{3} \cos 3x \right) + C$ $= -\frac{1}{26} \cos 13x - \frac{1}{6} \cos 3x + C$	2 Marks Correct solution 1 Mark Obtains $\frac{1}{2} \int (\sin 13x + \sin 3x) \, dx$
Q13b	$V = \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{2} (1 + \sin x) \right)^2 dx$ $V = \frac{\pi}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + 2 \sin x + \sin^2 x) dx$ $V = \frac{\pi}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(1 + 2 \sin x + \frac{1}{2} (1 - \cos 2x) \right) dx$ $V = \frac{\pi}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{3}{2} + 2 \sin x - \frac{1}{2} \cos 2x \right) dx$ $V = \frac{\pi}{4} \left[\frac{3}{2} x - 2 \cos x - \frac{1}{4} \sin 2x \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$ $V = \frac{\pi}{4} \left[\left(\frac{3}{2} \times \frac{\pi}{2} - 2 \cos \frac{\pi}{2} - \frac{1}{4} \sin \left(2 \times \frac{\pi}{2} \right) \right) - \left(\frac{3}{2} \times -\frac{\pi}{2} - 2 \cos \left(-\frac{\pi}{2} \right) - \frac{1}{4} \sin \left(2 \times -\frac{\pi}{2} \right) \right) \right]$ $V = \frac{\pi}{4} \left[\left(\frac{3\pi}{4} - 0 - 0 \right) - \left(-\frac{3\pi}{4} - 0 \right) \right]$ $V = \frac{3\pi^2}{8} \text{ units}^3$	3 Marks Correct solution 2 Marks Correct anti-derivative 1 Mark Finds $\frac{\pi}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + 2 \sin x + \sin^2 x) dx$

Q13ci	$\overrightarrow{XM} = \overrightarrow{XB} + \overrightarrow{BM}$ $\overrightarrow{XM} = \frac{3}{4}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{BD}$ $\overrightarrow{XM} = \frac{3}{4}\overrightarrow{DC} + \frac{1}{2}(\overrightarrow{BA} + \overrightarrow{AD})$ $\overrightarrow{XM} = \frac{3}{4}(8\mathbf{a} + 4\mathbf{b}) + \frac{1}{2}(-8\mathbf{a} - 4\mathbf{b} - 12\mathbf{a} - 20\mathbf{b})$ $\overrightarrow{XM} = (6\mathbf{a} + 3\mathbf{b}) + (-10\mathbf{a} - 12\mathbf{b})$ $\overrightarrow{XM} = -4\mathbf{a} - 9\mathbf{b}$	2 Marks Correct solution 1 Mark Finds $\overrightarrow{XB}$ or $\overrightarrow{BM}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Q13cii	$\overrightarrow{MY} = \overrightarrow{MB} + \overrightarrow{BY}$ $\overrightarrow{MY} = \overrightarrow{MB} + \overrightarrow{BC} + \overrightarrow{CY}$ $\overrightarrow{MY} = -\overrightarrow{BM} - \overrightarrow{CB} - \frac{1}{2}\overrightarrow{DA}$ $\overrightarrow{MY} = (10\mathbf{a} + 12\mathbf{b}) - (12\mathbf{a} + 20\mathbf{b}) - \frac{1}{2}(12\mathbf{a} + 20\mathbf{b})$ $\overrightarrow{MY} = 10\mathbf{a} + 12\mathbf{b} - 12\mathbf{a} - 20\mathbf{b} - 6\mathbf{a} - 10\mathbf{b}$ $\overrightarrow{MY} = -8\mathbf{a} - 18\mathbf{b}$ $\overrightarrow{XY} = \overrightarrow{XB} + \overrightarrow{BY}$ $\overrightarrow{XY} = 6\mathbf{a} + 3\mathbf{b} - 12\mathbf{a} - 20\mathbf{b} - 6\mathbf{a} - 10\mathbf{b}$ $\overrightarrow{XY} = -12\mathbf{a} - 27\mathbf{b}$ $\overrightarrow{XY} = 3\overrightarrow{XM} = \frac{3}{2}\overrightarrow{MY}$ Since these vectors are scalar multiples of each other and M is a common point, therefore X, M and Y are collinear. $k = \frac{1}{2}$	3 Marks Correct solution 2 Marks Finds k or proves collinear 1 Mark Finds $\overrightarrow{MY}$ or $\overrightarrow{XY}$
Q13di	$\frac{x+2}{4-x^2}$ $= \frac{x+2}{(2+x)(2-x)}$ $= \frac{1}{2-x}$	1 Mark Correct solution
Q13dii	$2(4-x^2)\frac{dy}{dx} = y(x+2)$ $\int \frac{2}{y} dy = \int \frac{x+2}{4-x^2} dx$ $\int \frac{2}{y} dy = \int \frac{1}{2-x} dx$ $2\ln y = -\ln 2-x + C_1$ $\ln y = -\frac{1}{2}\ln 2-x + C_2$ $\ln y = \ln \frac{1}{\sqrt{2-x}} + C_2$ $y = Ae^{\frac{\ln \frac{1}{\sqrt{2-x}}}{1}}$ $y = \frac{A}{\sqrt{2-x}}$ $x=1, y=2$ $2 = \frac{A}{\sqrt{2-1}}$ $A=2$ $\therefore y = \frac{2}{\sqrt{2-x}}$	3 Marks Correct solution 2 Marks Obtains correct primitive 1 Mark Separates the variables in the differential equation, or equivalent merit

Q14a		3 Marks Correct solution 2 Marks Correct graph with most key features shown 1 Mark Identifies some features and one correct branch
Q14bi	$\frac{dN}{dt} = kN \left(1 - \frac{N}{5000} \right)$ $500 = k \times 1000 \times \left(1 - \frac{1000}{5000} \right)$ $500 = k \times 800$ $k = \frac{5}{8} = 0.625$	1 Mark Correct solution
Q14bii	$\frac{1}{N} + \frac{1}{5000 - N}$ $= \frac{5000 - N + N}{N(5000 - N)}$ $= \frac{5000}{N(5000 - N)}$	1 Mark Correct solution
Q14biii	$\frac{dN}{dt} = 0.625N \left(1 - \frac{N}{5000} \right)$ $\frac{dN}{dt} = 0.625N \left(\frac{5000 - N}{5000} \right)$ $\int \frac{5000}{N(5000 - N)} dN = \int 0.625 dt$ $\int \left(\frac{1}{N} + \frac{1}{5000 - N} \right) dN = \int 0.625 dt$ $\ln N - \ln 5000 - N = 0.625t + C$ $\frac{N}{5000 - N} = Ae^{0.625t}$ $t = 0, N = 200$ $\frac{200}{5000 - 200} = Ae^0$ $A = \frac{1}{24}$ $\frac{N}{5000 - N} = \frac{1}{24} e^{0.625t}$ $24N = e^{0.625t}(5000 - N)$ $24N + e^{0.625t}N = 5000e^{0.625t}$ $N(24 + e^{0.625t}) = 5000e^{0.625t}$ $N = \frac{5000e^{0.625t}}{24 + e^{0.625t}}$ $t = 6$ $N = \frac{5000e^{0.625 \times 6}}{24 + e^{0.625 \times 6}}$ $N = 3196.06 \dots$ $N = 3196$ The number of people in the arena after 6 minutes is approximately 3196.	4 Marks Correct solution 3 Marks Makes significant progress and rearranging the equation to find N 2 Marks Integrate both sides correctly 1 Mark Separating the variables in the differential equation

Q14biv	$90\% \times 5000 = 4500$ $\frac{4500}{5000 - 4500} = \frac{1}{24} e^{0.625t}$ $9 = \frac{1}{24} e^{0.625t}$ $9 \times 24 = e^{0.625t}$ $\ln 216 = 0.625t$ $t = \frac{\ln 216}{0.625}$ $t = 8^{\circ}36'1.6''$ $t = 8 \text{ min } 36 \text{ s (nearest second)}$	1 Mark Correct solution
Q14ci	Let $\sin^{-1} x = \alpha, \cos^{-1} x = \beta$ $\sin \alpha = \frac{x}{1}, \cos \beta = \frac{x}{1}$  $\sin(\sin^{-1} x - \cos^{-1} x)$ $= \sin(\alpha - \beta)$ $= \sin \alpha \cos \beta - \sin \beta \cos \alpha$ $= \frac{x}{1} \times \frac{x}{1} - \frac{\sqrt{1-x^2}}{1} \times \frac{\sqrt{1-x^2}}{1}$ $= x^2 - (1 - x^2)$ $= 2x^2 - 1$	3 Marks Correct solution 2 Marks Makes significant progress 1 Mark Identifies $\sin \beta = \cos \alpha$ $= \sqrt{1 - x^2}$
Q14cii	$\sin^{-1} x - \cos^{-1} x = \sin^{-1}(2 - x)$ $\sin(\sin^{-1} x - \cos^{-1} x) = 2 - x$ $2x^2 - 1 = 2 - x$ $2x^2 + x - 3 = 0$ $2x^2 - 2x + 3x - 3 = 0$ $2x(x - 1) + 3(x - 1) = 0$ $(x - 1)(2x + 3) = 0$ $x = 1, x = -\frac{3}{2}$ $0 \leq x \leq \frac{\pi}{2}$ $\therefore x = 1 \text{ is the only solution.}$	2 Marks Correct solution 1 Mark Deduce $2x^2 - 1 = 2 - x$ and find both x values